
Task #2 - Outlier identification

Description

Mining for outliers is of great importance in many domains: fraud detection, disease identification, intrusion detection, fault diagnosis, etc...

Outliers or **anomalies** represent rare observations/events that deviate from the majority of data either in magnitude or with respect to an overall pattern. Whatever the data, a statistical model can be attached to it and consequently the data can be considered to be generated by a statistical process. In this view, outliers correspond to very low probabilities under the underlying distribution.

Outliers may sometimes simply occur due to erroneous recording of data and not due to meaningful anomalous data which would correspond in the theoretical model to changes in the generative process. Identifying them must be **one of the first steps in data analysis** as their presence misleads many algorithms from the machine learning area solving tasks like clustering, classification or regression.

Traditional methods for outlier detection generally belong to three classes [1]:

- **Statistical methods** for outlier detection are mostly parametric methods: given a certain kind of statistical distribution, outliers are detected as those points with low probability of being generated. In the *univariate* case usually the Normal distribution is used and outliers are considered those observations that lie at a distance larger than k *standard deviations* from the mean. For the multivariate case the Mahalanobis distance is considered to compute the distance of the observations from the mean. The main drawback of this approach is that the parameters of the distribution (mean and standard deviations) are computed based on all observations, including the possible outliers, and therefore they may be highly biased. Non-parametric approaches based on standard deviation identify the subset of observations that, after exclusion, determines the highest decrease in variance; using any form of the Minkowski metric the method can be generalized for the multivariate case.
- **Distance-based approaches** for outlier detection base their decisions on computing the distances between each data point and its neighbors. These are multivariate approaches. The main advantage of these methods over

statistical approaches is that no hypothesis on the type of distribution is made, hence such methods are applicable without distribution-dependent restrictions. These methods are computationally expensive due to the calculations of distances between data points; therefore, various ways of scaling them up for large databases have been proposed. Still, one important drawback is present: even if they do not make any assumptions on the type of distribution they are not parametric-free methods. The result is very sensitive to some user-tuned parameters like the radius of the neighborhood, the number of neighbors to be used or the threshold indicating the average distance to the neighbors above which a data is considered as outlier.

- **Clustering-based approaches** employ an unsupervised clustering algorithm to identify groups in data. In this context outliers are identified as unusually small groups in data. Popular algorithms used in this context are hierarchical clustering methods (mostly the single-link variant) and density based clustering methods like DBSCAN. The methods have the advantage of being generally applicable to any distributions in data. Their drawbacks reside in high computational costs requiring distance computations between data items and sensitivity to parameters - in case of density-based methods.

Modern techniques are based on neural networks (performing a type of factor analysis).

Use the two univariate and four of the multivariate techniques from the list below.

Analyze the impact of data standardization (Z score normalization).

All the multivariate methods output a numerical score (may be a distance) for each data item; this will be analyzed/plotted further in order to decide which items are most likely outliers in data. For each multivariate method, test **multiple score thresholds** for classifying points as outliers.

The first task is to choose one dataset from [this list](#) and apply the indicated methods for outlier detection. Compute **precision** and **recall** for each method you use.

The second task is to identify outliers on your data set. Refer back to the univariate and multivariate plots and also apply the 6 required techniques. Compare them by plotting the resulting thresholds and **compute an intersection between all the different algorithms**.

Details

- Univariate techniques:
 - Mean $\pm k \cdot sd$
 - [1.5 IQR](#)
- Multivariate techniques:
 - [Mahalanobis distance](#)
 - [LOF](#)
 - Projection methods: [stahel-donoho](#)
 - [Autoencoders](#)
 - [Isolation forest](#)

Resources

- https://scikit-learn.org/stable/modules/outlier_detection.html
- <https://statisticsbyjim.com/basics/outliers/>
- https://docs.oracle.com/cd/E17236_01/epm.1112/cb_statistical/frameset.htm
- <https://www.analyticsvidhya.com/blog/2021/05/feature-engineering-how-to-detect-and-remove-outliers-with-python-code/>

Grading

Requirement	Poin ts
Task 1- work with labeled data and report precision and recall for each method - 2 univariate methods (1p) - 4 multivariate methods (2p) - Data standardization only for methods for which it is needed (0.4p) - Visualizing data/extracted outliers (1.6p) - Identifying good thresholds in the multivariate case (2p)	7p
Task 2 - Identifying outliers on your dataset, using the same methods from Task1 (2.5p) - Intersection between different algorithms (0.5p)	3p

References

- [1] Mihaela Breaban, and Henri Luchian. "Outlier detection with nonlinear projection pursuit." *International Journal of Computers Communications & Control* 8.1 (2012): 30-36.
<http://univagora.ro/jour/index.php/ijccc/article/download/165/4>